



Chapter # 1

Numerical:

1. What is the percent uncertainty in the measurement 3.67 ± 0.25 mm?

Data:

Measurement = 3.67 ± 0.25 mm

Percent uncertainty = ?

Solution:

$$\text{Percent Uncertainty} = \frac{\text{absolute uncertainty}}{\text{value measure}} \times 100$$

$$\text{Percent Uncertainty} = \frac{0.25}{3.67} \times 100$$

$$\text{Percent Uncertainty} = 6.8 \%$$

2. What is the area, and its approximate uncertainty, of a circle with radius 3.7×10^4 cm. The uncertainty in the radius is given as 0.05×10^4 cm

Data:

$A = ?$

Uncertainty = $\Delta A = ?$

$r = 3.7 \times 10^4$ cm.

Solution:

$$A = \pi r^2$$

$$A = \pi (3.7 \times 10^4)^2$$

$$A = 4.3 \times 10^9 \text{ cm}^2$$

$$\Delta A = 2\pi r \Delta r$$

$$\Delta A = 2\pi (3.7 \times 10^4) (0.05 \times 10^4)$$

$$\Delta A = 0.11 \times 10^9 \text{ cm}^2$$

3. An aero plane travels at 850 km/h. How long does it take to travel 1.00 km?

Data:

$$v = 850 \text{ km/h} = 850 \times 1000 \div 3600 = 236.111 \text{ m/s}$$

$$s = 1.00 \text{ km} = 1 \times 1000 = 1000 \text{ m}$$

Solution:

$$v = \frac{s}{t}$$

$$t = \frac{s}{v}$$

$$t = \frac{1000}{236.111}$$

$$t = 4.23529 \text{ s}$$

Since the given distance value (1.00 km) has three significant figures, the calculated time should also be expressed with three significant figures.



$$t = 4.2353$$

$$t = 4.235$$

$$t = 4.24 \text{ s}$$

4. A rectangular holding tank 25.0 m in length and 15.0 m in width is used to store water for short period of time in an industrial plant. If 2980m³ water is pumped into the tank. What is the depth of the water?

Data:

$$L = 25.0 \text{ m}$$

$$W = 15.0 \text{ m}$$

$$V = 2980\text{m}^3$$

$$H = ?$$

Solution:

$$V = L \times W \times H$$

$$H = \frac{V}{L \times W}$$

$$H = \frac{2980}{25 \times 15}$$

$$H = 7.9466 \text{ m}$$

$$H = 7.947 \text{ m}$$

$$H = 7.95 \text{ m}$$

5. Find the volume of rectangular underground water tank has storage facility of 1.9 m by 1.2 m by 0.8 m.

Data:

$$L = 1.9 \text{ m}$$

$$W = 1.2 \text{ m}$$

$$H = 0.8 \text{ m}$$

Solution:

$$V = L \times W \times H$$

$$V = 1.9 \times 1.2 \times 0.8$$

$$V = 1.824 \text{ m}^3$$

We need to use the lowest number of significant figures among the given dimensions, which is 1 significant figure. Therefore, we should round the volume to 1 significant figure.

$$V = 2 \text{ m}^3$$

6. Two students derive following equations in which x refers to distance travelled, v the speed, a the acceleration, and t the time and the subscript (o) means a quantity at time t=0

a) $x = v t^2 + 2at$

b) $x = v_o t + 2at^2$

Which of these could possibly be correct according to dimensional check?



a) $[L] = [L].[T]^{-1} \cdot [T]^2 + 2 [L].[T]^{-2} \cdot [T]$

$[L] = [L].[T] + 2 [L].[T]^{-1}$

$[L] = [L].[T] + 2 [L].[T]^{-1}$

Since L.H.S $\neq$ R.H.S, it is incorrect according to dimension.

b) $[L] = [L].[T]^{-1} \cdot [T] + 2 [L].[T]^{-2} \cdot [T]^2$

$[L] = 3[L]$

After omitting constant

$[L] = [L]$

Since L.H.S = R.H.S, it is correct according to dimension.

7. One hectare is defined as 104 m². One acre is 4 x 10⁴ ft². How many acres are in one hectare? (Hint: 1 m = 3.28 ft).

Data:

Hectare = 10⁴ m²

Acre = 4 x 10⁴ ft²

Acres in one hectare = ?

1 m = 3.28 ft

Solution:

Area in square feet= 10⁴ x 3.28

Area in square feet= 10⁴ x 10.76

Area in square feet = 107600 ft²

Number of acres = $\frac{107600}{4 \times 10^4}$

Number of acres = $\frac{\text{Area in square feet}}{4 \times 10^4}$

Number of acres = 2.69

8. A watch factory claims that its watches gain or lose not more than 10 seconds in a year. How accurate is this watch, express as percentage?

Data:

Acceptable range = 10 s

Percentage Accuracy = ?

Solution:

Percentage Accuracy = $\frac{\text{Acceptable Range}}{\text{Total Seconds in a year}} \times 100$

Percentage Accuracy = $\frac{10}{365 \times 24 \times 60 \times 60} \times 100$

Percentage Accuracy = 3.17 x 10⁻⁵

9. The diameter of Moon is 3480 km. What is the volume of the Moon? How many Moons would be needed to create a volume equal to that of Earth?



Data:

$$d = 3480 \text{ km}$$

$$V = ?$$

$$\text{No. of Moons} = ?$$

Solution:

First we calculate the volume of moon.

$$r = \frac{d}{2} = \frac{3480}{2} = 1740 \text{ km} = 1740 \times 10^3 \text{ m}$$

$$V = \pi r^3$$

$$V = \pi (1740 \times 10^3)^3$$

$$V = 2.2 \times 10^{19} \text{ m}^3$$

Now, we calculate the no. of moons:

$$\text{No. of Moons} = \frac{\text{volume of earth}}{\text{volume of moon}}$$

$$\text{Volume of Earth} = 1.08321 \times 10^{21} \text{ km}^3$$

$$\text{Volume of Earth} = 1.08321 \times 10^{21} \text{ m}^3$$

$$\text{No. of Moons} = \frac{1.08321 \times 10^{21}}{2.2 \times 10^{19}}$$

$$\text{No. of Moons} = 49.24$$

**Chapter # 2****Numerical:**

1. A helicopter is ascending at the rate of 12 m/s. At a height of 80 m above the ground, a package is dropped. How long does the package take to reach the ground?

Solution:

Consider upward motion from B to C.

Initial velocity of package at B = $v_i = 12$ m/sec

Final velocity of package at C = $v_f = 0$

Time from B to C = $t_1 = ?$

$$a = -g = -9.8 \text{ m/s}^2$$

Using the equation of motion

$$v_f = v_i + at$$

$$0 = 12 + (-9.8) t_1$$

$$9.8 t_1 = 12$$

$$t_1 = 1.2 \text{ sec}$$

To calculate distance from B to C

$$S_1 = v_i t + \frac{1}{2} g t^2$$

$$S_1 = (12)(1.2) + \frac{1}{2} (9.8) (1.2)^2$$

$$S_1 = 14.4 - 7.06$$

$$S_1 = 7.34 \text{ m}$$

Consider downward motion from C to A

$$S = 80 + S_1$$

$$S = 80 + 7.34 = 87.34 \text{ m}$$

$$v_i = 0$$

$$g = 9.8 \text{ m/sec}^2$$

$$t_2 = ?$$

using equation of motion

$$S = v_i t + \frac{1}{2} g t^2$$

$$87.34 = 0 \times t_2 + (9.8) t_2^2$$

$$87.34 = 4.9 t_2^2$$

$$t_2 = 4.22 \text{ sec}$$

Hence total time the package take to reach the ground is

$$t = t_1 + t_2$$

$$t = 1.2 + 4.22$$

$$t = 5.42 \text{ sec}$$

2. Two tug boats are towing a ship each exerts a force of 6000 N, and the angle between two ropes is 60°. Calculate the resultant force on the ship?



Data:

$$F_1 = F_2 = 6000 \text{ N}$$

$$\theta = 60^\circ$$

$$F = ?$$

Solution:

$$F = \sqrt{F_x^2 + F_y^2 + 2F_xF_y\cos\theta}$$

$$F = \sqrt{6000^2 + 6000^2 + 2(6000)(6000)\cos 60}$$

$$F = 10392.3 \text{ N}$$

3. A car starts from rest and moves with a constant acceleration. During the 5th second of its motion, it covers a distance of 36 meters. Calculate:

a) acceleration of the car

b) the total distance covered by the car during this time

Solution:

Distance covered by the car in 5 second = $S_5 = ?$

$$S_5 = S_4 + 36$$

Initial velocity of the car = $v_i = 0$

Time = $t = 5 \text{ sec}$

$$a = ?$$

Using equation of motion

$$S = vit + \frac{1}{2}at^2$$

$$S_5 = 0 \times 5 + \frac{1}{2}(a)(5)^2$$

$$S_5 = 12.5a$$

Distance covered by the car in 4 second = S_4

Initial velocity of the car = $v_i = 0$

Time = $t = 4 \text{ sec}$

$$a = ?$$

Using equation of motion

$$S = vit + \frac{1}{2}at^2$$

$$S_4 = 0 \times 4 + (a)(4)^2$$

$$S_4 = 8a$$

Now the distance covered by the minibus in 5th second.

$$S_5 = S_4 + 36$$

$$S_5 - S_4 = 36$$

$$12.5a - 8a = 36$$

$$a = 8 \text{ m/s}^2$$

Putting the value of a in S_5

$$S_5 = 12.5(8) = 100 \text{ m}$$



4. Show that the range of projectile at complementary angles is same with examples?

Solution:

Two angles are complementary if

$$\alpha + \beta = 90^\circ$$

$$\text{Let } \alpha = 30^\circ \text{ and } \beta = 60^\circ$$

$$R = \frac{v_o^2 \sin 2\alpha}{g}$$

$$R = \frac{v_o^2 \sin 2 \times 30}{g}$$

$$R = 0.866 \frac{v_o^2}{g}$$

$$R' = \frac{v_o^2 \sin 2\beta}{g}$$

$$R' = \frac{v_o^2 \sin 2 \times 60}{g}$$

$$R' = 0.866 \frac{v_o^2}{g}$$

Therefore, $R = R'$

5. At what angle the range of projectile becomes equal to the height of projectile?

Solution:

$$H = R$$

$$\frac{v_o^2 \sin^2 \theta}{2g} = v_o^2 \sin 2\theta$$

$$\frac{\sin^2 \theta}{2} = 2 \sin \theta \cos \theta$$

$$\frac{\sin \theta}{2} = 2 \cos \theta$$

$$\frac{\sin \theta}{\cos \theta} = 4$$

$$\tan \theta = 4$$

$$\theta = \tan^{-1} 4$$

$$\theta = 76^\circ$$

6. A mortar shell is fired at a ground level target 500 m distance with an initial velocity of 90 m/s. What is the launch angle?

Data:

$$R = 500 \text{ m}$$

$$v_o = 90 \text{ m/s}$$

$$\theta = ?$$

Solution:

$$R = \frac{v_o^2 \sin 2\alpha}{g}$$

$$500 = \frac{90^2 \sin 2\theta}{9.8}$$

$$\frac{500 \times 9.8}{8100} = \sin 2\theta$$



$$\sin 2\theta = 0.604$$

$$2\theta = \sin^{-1}(0.604)$$

$$2\theta = 37.22$$

$$\theta = 18.61^\circ$$

If the shell is fired with angle ($90 - 18.6$) = 71.4° then range will also 500m.

**Chapter # 3****Numericals:**

1. A car weighing 9800 N is moving with a speed of 40 Km/h. On the Application of breaks it comes to rest after traveling a distance of 50 meters. Calculate the average retarding force?

Data:

$$W = 9800 \text{ N}$$

$$v = 40 \text{ Km/h} = 40 \times 1000 \div 3600 = 11.111 \text{ m/s}^2$$

$$S = 50 \text{ m}$$

$$F = ?$$

Solution:

$$F = ma$$

$$m = \frac{w}{g}$$

$$m = \frac{9800}{9.8}$$

$$m = 1000 \text{ kg}$$

$$2aS = vf^2 - vi^2$$

$$2a(50) = 0^2 - 11.111^2$$

$$100a = -123.43$$

$$a = -1.234 \text{ m/s}^2$$

The negative sign shows that the velocity of the car is decreasing.

$$a = -1.2345 \text{ m/s}^2$$

$$F = ma$$

$$F = (1000) (-1.2345) = -1234.5 \text{ N.}$$

2. A helicopter weighs 3920 N. Calculate the force on it if it is ascending up at the rate of 2m/s². What will be the force on helicopter if it is moving up with a constant speed of 4 m/s?

Data:

$$W=3920 \text{ N}$$

$$a = 2 \text{ m/sec}^2$$

$$v = 4 \text{ m/sec}$$

$$F = ?$$

$$F' = ?$$

Solution:

$$\text{Mass of helicopter} = m = \frac{w}{g}$$

$$m = 400 \text{ kg}$$



Since helicopter is ascending up therefore the net force on the helicopter is

$$F - W = ma$$

$$F = ma + W$$

$$F = 3920 + 400 \times 2$$

$$F = 3920 + 800$$

$$\mathbf{F = 4720\ N}$$

Force on of helicopter when ascending with constant velocity. In this case acceleration of helicopter is zero

$$F - W = ma$$

$$F - W = m(0)$$

$$F - W = 0$$

$$F = W$$

$$\mathbf{F = 3920\ N}$$

3. A 100 grams bullet s fired from a 10 kg gun with a speed of 100m/s. What is the speed of recoil of the gun?

Data:

$$m_1 = 100\ g = = 0.1\ kg$$

$$m_2 = 10\ kg$$

$$\text{Velocity of bullet before firing} = u_1 = 0$$

$$\text{Velocity of gun before firing} = u_2 = 0$$

$$\text{Velocity of bullet after firing} = v_1 = 1000\ m/s$$

$$\text{Velocity of gun after firing} = v_2 = ?$$

Solution:

According to the law of conservation of linear momentum

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$(0.1)(0) + (10)(0) = (0.1)(1000) + (10) v_2$$

$$0+0=100+10v_2$$

$$10v_2 = -100$$

$$\mathbf{v_2 = -10\ m/s}$$

Negative sign shows that the direction of velocity of recoil of gun is opposite to the direction of velocity of bullet.

4. A machine gun fires 10 bullets per second into a target. Each bullet weighs 20 gram and had a speed of 1500 m/s. Find the necessary force to hold the gun in position?

Data:

$$\text{Mass of bullet} = m = 20g = = 0.02\ kg$$

$$\text{Number of bullet} = 10$$

$$\text{Mass of 10 bullets} = M = 10 \times 0.02 = 0.2\ kg$$

$$\text{Time} = t = 1s$$



Initial velocity of bullet = $v_i = 0$

Final velocity of bullet = $v_f = 1500$ m/sec

Solution:

Force necessary to hold the gun = change in momentum of bullets per second

$$F = \frac{m(v_f - v_i)}{t}$$

$$F = \frac{0.02(1500 - 0)}{1}$$

$$F = 300 \text{ N}$$

5. A 50 grams bullet is fired into a 10 kg block that is suspended by a long cord so that it can swing as a pendulum. If the block is displaced so that its center of gravity rises by 10cm, what was the speed of bullet?

Data:

$$m_1 = 50\text{g}$$

$$m_1 = 0.05\text{Kg}$$

$$m_2 = 10\text{Kg}$$

$$h = 10 \text{ cm} = 0.1 \text{ m}$$

$$\text{Velocity of block before impact} = u_2 = 0$$

$$\text{Velocity of bullet after impact} = v_1 = V$$

$$\text{Velocity of block after impact} = V_2 = V$$

$$\text{Velocity of bullet before impact} = u_1 = ?$$

Solution:

According to law of conservation of linear momentum

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$(0.05) (u_1) + (10) (0) = 0.05 v + 10 v$$

$$0.05 u_1 = 10.05 v$$

$$u_1 = v$$

$$u_1 = 201 v$$

According law of conservation of energy

loss of K.E = gain of P.E

$$\frac{1}{2} m v^2 = m g h$$

$$v^2 = 2 g h$$

$$v = \sqrt{2 g h}$$

$$v = \sqrt{2(9.8)(0.1)}$$

$$v = 1.4 \text{ m/s}$$

Putting the value of v is eq (1)

$$u_1 = 201 \times 1.4 = 281.4 \text{ m/s}$$

6. A 70-gram ball collides with another ball of mass 140 gram. The initial velocity of the first ball is 9m/s to right while the second ball is at rest. If the



collision were perfectly elastic, what would be the velocity of two balls after the collision?

Data:

$$m_1 = 70 \text{ gm} = 0.07 \text{ kg}$$

$$m_2 = 140 \text{ gm} = 0.14 \text{ kg}$$

$$\text{Initial velocity of first ball} = U_1 = 9 \text{ m/s}$$

$$\text{Initial velocity of second ball} = U_2 = 0$$

$$\text{Final velocity of first ball} = V_1 = ?$$

$$\text{Final velocity of second ball} = V_2 = ?$$

Solution:

Final velocity of first ball after collision:

$$V_1 = \frac{(m_1 - m_2)U_1}{m_1 + m_2} + \frac{2m_2U_2}{m_1 + m_2}$$

$$V_1 = \frac{(0.07 - 0.14)9}{0.7 + 0.14} + \frac{2(0.7)(0.14)}{0.7 + 0.14}$$

$$V_1 = -3 \text{ m/s}$$

Here -ve sign shows that first ball will return back in opposite direction after collision.

Find Velocity of steel ball after collision:

$$V_2 = \frac{(m_2 - m_1)U_2}{m_1 + m_2} + \frac{2m_1U_1}{m_1 + m_2}$$

$$V_2 = \frac{(0.14 - 0.07)0}{0.14 + 0.07} + \frac{2(0.07)(9)}{0.14 + 0.07}$$

$$V_2 = 6 \text{ m/s}$$

7. A truck weighing 2500 kg and moving with a velocity of 21 m/s collides with a stationary car weighing 1000 kg. The truck and car move together after the impact. Calculate their common velocity?

Data:

$$m_1 = 2500 \text{ kg}$$

$$m_2 = 1000 \text{ kg}$$

$$u_1 = 21 \text{ m/s}$$

$$u_2 = 0$$

$$v_1 = v_2 = v = ?$$

Solution:

$$(2500)(21) + (1000)(0) = (2500)(v) + (1000)v$$

$$v = 15 \text{ m/s}$$

**Chapter # 4****Numericals:**

1. A car mechanic applies a force of 800 N to a wrench for the purpose of loosening a bolt. He applies the force which is perpendicular to the arm of the wrench. The distance from the bolt to the mechanic's hand is 0.40 m. Find out the magnitude of the torque applied?

Data:

$$F = 800 \text{ N}$$

$$\theta = 90^\circ \text{ (applies the force which is perpendicular to the arm of the wrench)}$$

$$r = 0.40 \text{ m}$$

$$\tau = ?$$

Solution:

$$\tau = F \times r$$

$$\tau = F r \sin\theta$$

$$\tau = (800) (0.40) \sin 90$$

$$\tau = 320 \text{ N.m}$$

2. A car accelerates uniformly from rest and reaches a speed of 22 m/s in 9s. If the diameter of a tire is 58 cm, find (a) The number of revolutions the tire makes during this motion, assuming no slipping and (b) The final rotational speed of the tire in revolutions per second.

Data:

$$V_i = 0$$

$$V_f = 22 \text{ m/s}$$

$$t = 9 \text{ s}$$

$$d = 58 \text{ cm} = 58 \div 100 = 0.58 \text{ m}$$

$$(a) N = ?$$

$$(b) \omega_f = ?$$

Solution:

(a)

$$V_f = V_i + at$$

$$22 = 0 + a (9)$$

$$a = 2.444 \text{ m/s}^2$$

$$S = (0) (9) + (2.444) (9)^2$$

$$S = 99 \text{ m}$$

$$\text{Circumference of tire} = \pi d = (\pi) (0.58) = 1.822 \text{ m}$$

$$\text{So in one revolution it covers 1.822 m, therefore it will make} = 54.335 \text{ rev} \approx 55 \text{ rev}$$

(b)

$$a = r \alpha$$

$$r = 0.29 \text{ m}$$



$$a = r \alpha$$

$$2.444 = (0.29) \alpha$$

$$\alpha = 2.444 \div 0.29$$

$$\alpha = 8.43 \text{ rad/s}^2$$

$$\omega_f = \omega_i + \alpha t$$

$$\omega_f = 0 + (8.43) (9) = 75.87 \text{ rad/s}$$

$$\omega_f = 75.87 \div 2\pi = 12 \text{ rev/s}$$

3. An ordinary workshop grindstone has a radius of 7.5 cm and rotates 6500 rev/min. (a) Calculate the magnitude of centripetal acceleration at its edge in m/s² and convert it into multiples of g. (b) What is the linear speed of a point on its edge?

Data:

$$r = 7.5 \text{ cm} = 0.075 \text{ m}$$

$$\omega = 6500 \text{ rev/min} = 6500 \times 2\pi \div 60 = 680.678 \text{ rad/s}$$

$$(a) a_c = ?$$

$$a_c' = ?$$

$$(b) V = ?$$

Solution:

(a)

$$a_c = r \omega^2$$

$$a_c = (0.075) (680.678)^2$$

$$a_c = 3.47 \times 10^4 \text{ m/s}^2$$

Multiplying and dividing right hand side by g

$$a_c = 3.47 \times 10^4 \frac{g}{g}$$

$$a_c = 3.47 \times 10^4 \frac{g}{9.8}$$

$$a_c = 3.55 \times 10^3 g$$

(b)

$$V = r \omega$$

$$V = 0.075 \times 680.678$$

$$V = 51.05 \text{ m/s}$$

4. A satellite is orbiting the Earth with an orbital velocity of 3200 m/s. What is the orbital radius?

Data:

$$V = 3200 \text{ m/s}$$

$$R = ?$$

Solution:

$$V^2 = \frac{GM_{\text{central}}}{R}$$

$$3200^2 = \frac{6.67 \times 10^{-11} \times 6 \times 10^{24}}{R}$$



$$R = \frac{3200^2}{6.67 \times 10^{-11} \times 6 \times 10^{24}}$$

$$R = 3.897 \times 10^7 \text{ m}$$

5. A satellite wishes to orbit the earth at a height of 100 km (approximately 60 miles) above the surface of the earth. Determine the speed, acceleration and orbital period of the satellite. (Given: $M_{\text{earth}} = 5.98 \times 10^{24} \text{ kg}$, $R_{\text{earth}} = 6.37 \times 10^6 \text{ m}$)

Data:

$$h = 100 \text{ km} = 100 \times 1000 \text{ m} = 1 \times 10^5 \text{ m}$$

$$M_{\text{earth}} = 5.98 \times 10^{24} \text{ kg}$$

$$R_{\text{earth}} = 6.37 \times 10^6 \text{ m}$$

$$V = ?$$

$$a = ?$$

$$T = ?$$

Solution:

$$R = R_{\text{earth}} + h = 6.37 \times 10^6 + 1 \times 10^5 = 6.47 \times 10^6 \text{ m}$$

$$V = \sqrt{\frac{GM_{\text{central}}}{R}}$$

$$V = \sqrt{\frac{6.67 \times 10^{-11} \times 6 \times 10^{24}}{6.47 \times 10^6}}$$

$$V = 7.85 \times 10^3 \text{ m/s}$$

$$a = \frac{v^2}{r}$$

$$a = \frac{(7.85 \times 10^3)^2}{6.47 \times 10^6}$$

$$a = 9.524 \text{ m/s}^2$$

$$T^2 = \frac{4\pi^2 R^3}{GM_{\text{central}}}$$

$$T^2 = \frac{4\pi^2 (6.47 \times 10^6)^3}{6.67 \times 10^{-11} \times 6 \times 10^{24}}$$

$$T = 5177.04 \text{ s}$$

$$T = 5177.04 \div 60$$

$$T = 1.44 \text{ hours}$$

6. A thin disk with a 0.3m diameter and a total moment of inertia of 0.45 kg.m² is rotating about its center of mass. There are three rocks with masses of 0.2kg on the outer part of the disk. Find the total moment of inertia of the system?

Data:

$$d_{\text{disk}} = 0.3 \text{ m}$$

$$I_{\text{disk}} = 0.45 \text{ kg.m}^2$$

$$m_{\text{rock}} = 0.2 \text{ kg}$$

$$I_{\text{total}} = ?$$



Solution:

$$r_{\text{disk}} = \frac{d_{\text{disk}}}{2}$$

$$r_{\text{disk}} = \frac{0.3}{2} =$$

$$r_{\text{disk}} = 0.15 \text{ m}$$

$$I_{\text{rock}} = m_{\text{rock}} \cdot r_{\text{disk}}^2$$

$$I_{\text{rock}} = (0.2) (0.15)^2 = 4.5 \times 10^{-3} \text{ kg.m}^2$$

$$I_{\text{total}} = I_{\text{disk}} + 3I_{\text{rock}}$$

$$I_{\text{total}} = 0.45 + 3(4.5 \times 10^{-3})$$

$$I_{\text{total}} = \mathbf{0.464 \text{ kg.m}^2}$$

7. What is the ideal banking angle for a gentle turn of 1.20 km radius on a highway with a 105 km/h speed limit (about 65 mi/h), assuming everyone travels at the limit?

Data:

$$\theta = ?$$

$$r = 1.20 \text{ km} = 1.20 \times 1000 = 1200 \text{ m}$$

$$v = 105 \text{ km/h} = 105 \times 1000 \div 3600 = 29.167 \text{ m/s}$$

Solution:

$$\tan \theta = \frac{v^2}{rg}$$

$$\tan \theta = \frac{29.167^2}{1200 \times 9.8}$$

$$\theta = \mathbf{4.14^\circ}$$

8. A 1500-kg car moving on a flat, horizontal road negotiates a curve as shown in Figure. If the radius of the curve is 35.0 m and the coefficient of static friction between the tires and dry pavement is 0.523, find the maximum speed the car can have and still make the turn successfully.

Data:

$$m = 1500 \text{ kg}$$

$$r = 35.0 \text{ m}$$

$$\mu = 0.523$$

$$v = ?$$

Solution:

$$f = F_c$$

$$\mu mg = \frac{mv^2}{r}$$

$$(0.523)(1500)(9.8) = \frac{1500 v^2}{35}$$

$$v = \mathbf{13.4 \text{ m/s}}$$



9. A system of points shown in figure 4.22. Each particle has same mass of 0.3 kg and they all lie in the same plane. What is the moment of inertia of the



system about given axis?

Data:

$$m = 0.3 \text{ kg}$$

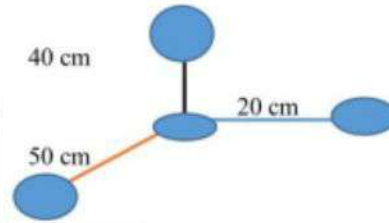
$$I = ?$$

Solution:

$$I = \sum m_i r_i^2 = m \sum r_i^2$$

$$I = 0.3 [0.5^2 + 0.4^2 + 0.2^2]$$

$$I = 0.135 \text{ kg.m}^2$$



10. (a) What is the angular momentum of a 2.9 kg uniform cylindrical grinding wheel of radius 20 cm when rotating 1550 rpm? (b) How much torque is required to stop it in 6s?

Data:

(a)

$$m = 2.9 \text{ kg}$$

$$r = 20 \text{ cm} = 20 \div 100 = 0.2 \text{ m}$$

$$\omega = 1550 \text{ rpm} = \frac{1550 \times 2\pi}{60} = 162.32 \text{ rad/s}$$

$$L = ?$$

(b)

$$\tau = ?$$

$$t = 6 \text{ s}$$

Solution:

(a)

$$L = I \omega$$

$$L = \left(\frac{1}{2} m r^2\right) \omega$$

$$L = \frac{1}{2} (2.9) (0.2)^2 (162.32)$$

$$L = 9.415 \text{ kg. m}^2/\text{s}$$

(b)

$$\tau = \frac{L_f - L_i}{t}$$

$$\text{Since } L_i = L = 9.415 \text{ kg. m}^2/\text{s}$$

$$\tau = \frac{0 - 9.415}{6}$$

$$\tau = -1.6 \text{ N.m}$$

11. Determine the angular momentum of the Earth (a) about its rotation axis (Assume the Earth as uniform sphere), and (b) in its orbit around the Sun (Take Earth as a particle orbiting the Sun). The Earth has mass $6 \times 10^{24} \text{ kg}$ and radius $6.4 \times 10^6 \text{ m}$, and is $1.5 \times 10^8 \text{ km}$ from the Sun.

Data:

$$M_e = 6 \times 10^{24} \text{ kg}$$



$$R = 6.4 \times 10^6 \text{ m}$$

$$R' = 1.5 \times 10^8 \text{ km} = 1.5 \times 10^{11} \text{ m}$$

(a)

$$I = ?$$

(b)

$$I' = ?$$

Solution:

(a)

$$I = \frac{2}{5} M_e R^2$$

$$I = \frac{2}{5} (6 \times 10^{24}) (6.4 \times 10^6)^2$$

$$I = 9.8 \times 10^{37} \text{ kg.m}^2$$

$$\omega = \frac{2\pi}{T}$$

$$T = 24 \times 60 \times 60 = 86400 \text{ s}$$

$$\omega = \frac{2\pi}{86400}$$

$$\omega = 7.27 \times 10^{-5} \text{ rad/s}$$

$$L = I \omega$$

$$L = (9.8 \times 10^{37}) (7.27 \times 10^{-5})$$

$$L = 7.125 \times 10^{33} \text{ kg.m}^2/\text{s}$$

(b)

$$I = M_e R'^2$$

$$I = (6 \times 10^{24}) (1.5 \times 10^{11})^2$$

$$I = 1.35 \times 10^{47} \text{ kg.m}^2$$

$$\omega = \frac{2\pi}{T}$$

$$T = 365 \times 24 \times 60 \times 60 = 31536000 \text{ s}$$

$$\omega = \frac{2\pi}{31536000} = 2 \times 10^{-7} \text{ rad/s}$$

$$L = I \omega = (1.35 \times 10^{47}) (2 \times 10^{-7}) = 2.7 \times 10^{40} \text{ kg.m}^2/\text{s}$$

**Chapter # 5****Numericals:**

1. A man pulls a trolley through a distance of 10 m by applying a force of 50 N which makes an angle of 60° with the horizontal. Calculate the work done by the man?

Data:

$$F = 50 \text{ N}$$

$$s = 10 \text{ m}$$

$$\theta = 60^\circ$$

$$W = ?$$

Solution:

$$W = Fs \cos \theta$$

$$W = (50) (10) \cos 60$$

$$W = 250 \text{ J}$$

2. A 100 kg man runs up a long flight of stairs in 9.8 second. The vertical height of the stair is 10 m. Calculate its power?

Data:

$$m = 100 \text{ kg}$$

$$t = 9.8 \text{ s}$$

$$h = 10 \text{ m}$$

$$P = ?$$

Solution:

$$P = \frac{W}{t}$$

$$\text{Since, } W = mgh$$

$$P = \frac{mgh}{t}$$

$$P = \frac{100 \times 9.8 \times 10}{9.8}$$

$$P = 1000 \text{ Watt}$$

3. When an object is thrown upwards. It rises to a height 'h'. How high is the object in terms of h? when it has lost one third of its original kinetic energy?

Data:

$$H = ?$$

$$K.E_{\text{final}} = K.E_{\text{initial}}$$

Solution:

$$mv_f^2 = mv_i^2$$

$$v_f^2 = v_i^2 \quad \text{eq (1)}$$

Now, if h is the maximum height achieved by the ball when $v_f=0$ then using the



kinematic equation give

$$K.E_{\text{final}} = K.E_{\text{initial}} - P.E_{\text{final}}$$

$$mv_f^2 = mv_i^2 - mgh$$

$$vf^2 = vi^2 - 2gh$$

$$0 = vi^2 - 2gh$$

$$vi^2 = 2gh$$

Now for the height H when $K.E_{\text{final}} = K.E_{\text{initial}}$,we use the kinematic equation using equation (1)

$$2gH = vi^2 + \frac{2}{3}vi^2$$

$$2gH = \frac{1}{3}vi^2$$

Putting vi^2

$$2gH = \frac{1}{3}2gh$$

$$H = \frac{h}{3}$$

4. A 70 kg man runs up a hill through a height of 3 m in 2 seconds. a) How much work does he do against gravitational field? b) What is the average power output?

Data:

$$m = 70 \text{ kg}$$

$$h = 3 \text{ m}$$

$$t = 2 \text{ s}$$

$$(a) W = ?$$

$$(b) P_{\text{av}} = ?$$

Solution:

(a)

$$W = mgh$$

$$W = (70) (9.8) (3)$$

$$W = 2058 \text{ J}$$

(b)

$$P_{\text{av}} = \frac{W}{t}$$

$$P_{\text{av}} = \frac{2058}{2}$$

$$P_{\text{av}} = 1029 \text{ W}$$

5. A neutron travels a distance of 12 m in a time interval of 3.6×10^{-4} sec. Assuming its speed was constant, Find its kinetic energy? Take the mass of neutron 1.7×10^{-27} kg.



Data:

$$s = 12\text{m}$$

$$t = 3.6 \times 10^{-4} \text{ s}$$

$$m = 1.7 \times 10^{-27} \text{ kg}$$

$$K.E = ?$$

Solution:

$$v = \frac{s}{t}$$

$$v = \frac{12}{3.6 \times 10^{-27}}$$

$$v = 33333.333 \text{ m/s}$$

$$K.E = \frac{1}{2} mv^2$$

$$K.E = \frac{1}{2} (1.7 \times 10^{-27}) (33333.333)^2$$

$$K.E = 9.444 \times 10^{-19} \text{ J}$$

$$K.E = 9.444 \times 10^{-19} \div (1.6 \times 10^{-19})$$

$$K.E = 5.895 \text{ eV}$$

6. A stone is thrown vertically upwards and can reach to a height of 10m, find the speed of stone, when it is just 2m above the ground?

Data:

$$h = 10\text{m}$$

$$h' = 2\text{m}$$

$$v' = ?$$

Solution:

$$2gh = v^2 - v_i^2$$

$$2(-9.8)(10) = 0 - v^2$$

$$v^2 = 196$$

$$\text{Initial K.E} + \text{Initial P.E} = \text{Final K.E} + \text{Final P.E}$$

$$\frac{1}{2} mv^2 + 0 = \frac{1}{2} mv'^2 + mgh'$$

$$\frac{1}{2} v^2 - gh' = \frac{1}{2} v'^2$$

$$\frac{1}{2} (196) - (9.8)(2) = \frac{1}{2} v'^2$$

$$v' = 12.52 \text{ m/s}$$

7. The potential energy of a body at the top of a building is 200 Joules when it is dropped its kinetic energy just before striking the ground is 160 Joules, find the work done against the air resistance?

Data:

$$P.E = 200 \text{ J}$$

$$K.E = 160 \text{ J}$$

$$W = ?$$

Solution:



$$K.E = P.E - W$$

$$160 = 200 - W$$

$$W = 200 - 160 = 40 \text{ J}$$

8. Find the energy equivalent of 1 gram?

Data:

$$m = 1 \text{ g} = 1 \div 1000 = 0.001 \text{ kg}$$

$$E = ?$$

Solution:

$$E = mc^2$$

$$E = (0.001) (3 \times 10^8)^2$$

$$E = 9 \times 10^{13} \text{ J}$$

9. A 1Kilowatt motor pump, pumps the water from the ground to a height of 10 m. Find, how much litres of water it can pump in one hour?

Data:

$$P = 1 \text{ kW} = 1000\text{W}$$

$$h = 10\text{m}$$

$$\frac{v}{t} = ?$$

Solution:

$$P = \frac{w}{t}$$

$$P = \frac{mgh}{t}$$

$$\frac{P}{gh} = \frac{m}{t}$$

$$\frac{m}{t} = \frac{1000}{9.8 \times 10}$$

$$\frac{m}{t} = 10.2 \text{ kg /s} \quad \text{eq(1)}$$

$$1 \text{ kg water} = 1 \text{ litre water}$$

$$\frac{m}{t} = \frac{v}{t}$$

eqn (1) becomes

$$\frac{v}{t} = 10.2$$

$$\frac{v}{t} = 10.2 \text{ litre/s}$$

$$\frac{v}{t} = 10.2 \times 3600$$

$$\frac{v}{t} = 36720 \text{ litre/h}$$

10. A rocket of mass 2kg is launched in air, when it attains height of 15m the 400 Joules of its chemical fuel burns. Find the speed of rocket at maximum height?



Data:

$$m = 2 \text{ kg}$$

$$h = 15 \text{ m}$$

$$W = 400 \text{ J}$$

$$V_f = ?$$

Solution:

$$\text{Gain in K.E} + \text{Gain in P.E} = W$$

$$\frac{1}{2} mv_f^2 + mgh = W$$

$$\frac{1}{2} (2) v_f^2 + (2) (9.8) (15) = 400$$

$$v_f = 10.3 \text{ m/s}$$

11. A motor pumps the water at the rate 500 gram/minute to the height of 120 m. If the motor is 50% efficient then how much input electric power is needed?

Data:

$$\frac{m}{t} = 500 \text{ gram/minute} = 500 \times 60 \div 1000 = 30 \text{ kg/s}$$

$$h = 120 \text{ m}$$

$$E = 50 \% = 50 \div 100 = 0.5$$

$$P_{\text{input}} = ?$$

Solution:

$$P_o = \frac{W}{t}$$

$$P_o = \frac{P.E}{t}$$

$$P_o = \frac{mgh}{t}$$

$$P_o = \frac{m}{t} gh$$

$$P_o = (30) (9.8) (120)$$

$$P_o = 35280 \text{ W}$$

$$E = \frac{\text{power input}}{\text{Power output}}$$

$$0.5 = \frac{P_i}{35280}$$

$$P_i = (35280) (0.5)$$

$$P_i = 17640 \text{ W}$$

OR

$$P_i = 17640 \div 746$$

$$P_i = 23.646 \text{ hp}$$

**Chapter # 6****Numericals:**

1. In a hydraulic press a force of 20 N is applied to a piston of area 0.20 m². The area of the other piston is 2.0 m². What is (a) the pressure transmitted through the fluid, (b) the force on the piston?

Data:

$$F_1 = 20 \text{ N}$$

$$A_1 = 0.20 \text{ m}^2$$

$$A_2 = 2.0 \text{ m}^2$$

$$(a) P = ?$$

$$(b) F_2 = ?$$

Solution:

(a)

$$P = \frac{F_1}{A_1} = \frac{20}{0.2}$$

$$P = 100 \text{ Pa}$$

(b)

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

OR

$$\frac{F_2}{F_1} = \frac{A_2}{A_1}$$

$$\frac{F_2}{20} = \frac{2}{0.2}$$

$$F_2 = 200 \text{ N}$$

2. The pressure in a water pipe in the ground floor of a building is $4 \times 10^5 \text{ Pa}$ but three floors up it is only $2 \times 10^5 \text{ Pa}$. What is the height between the ground floor and the third floor? The water in the pipe may be assumed to be stationary; density of water $= 1 \times 10^3 \text{ Kg/m}^3$.

Data:

$$P_1 = 4 \times 10^5 \text{ Pa}$$

$$P_2 = 2 \times 10^5 \text{ Pa}$$

$$h = ?$$

$$\rho = 1 \times 10^3 \text{ Kg/m}^3$$

$$g = 10 \text{ m/s}^2$$

Solution:

$$\Delta P = \rho gh$$

$$4 \times 10^5 - 2 \times 10^5 = (1 \times 10^3) (10) (h)$$

$$h = 20 \text{ m}$$



3. The small piston of hydraulic press has an area of 10.0 cm² If the applied force is 50.0 N. what must the area of the large piston to exert a pressing force of 4800N?

Data:

$$A_1 = 10.0 \text{ cm}^2$$

$$F_1 = 50.0 \text{ N}$$

$$F_2 = 4800 \text{ N}$$

$$A_2 = ?$$

Solution:

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

OR

$$\frac{F_2}{F_1} = \frac{A_2}{A_1}$$

$$\frac{4800}{500} = \frac{A_2}{10}$$

$$A_2 = 960 \text{ cm}^2$$

4. Mechanical advantage of a hydraulic jack is 420. Find the weight of the heaviest automobile that can be lifted by an applied force of 55 N.

Data:

$$M.A = 420$$

$$F_{in} = 55 \text{ N}$$

$$F_{out} = ?$$

Solution:

$$M.A = \frac{F_{out}}{F_{in}}$$

$$420 = \frac{F_{out}}{55}$$

$$F_{out} = 23100 \text{ N}$$

5. A flat-bottom river barge is 30 ft wide, 85 ft long and 15 ft deep. (a) how many m³ of water will displace while the top stays 1 m above the water? (b) What load in tons will the barge contain under these conditions if the empty barge weighs 160 tons in dry dock?

Data:

$$w = 30 \text{ ft} = 30 \div 3.28 = 9.15 \text{ m}$$

$$l = 85 \text{ ft} = 85 \div 3.28 = 26 \text{ m}$$

$$h = 15 \text{ ft} = 15 \div 3.28 = 4.6 \text{ m}$$

$$h' = 4.6 - 1 = 3.6 \text{ m}$$

$$m_{\text{empty}} = 160 \text{ tons} = 160 \times 1000 = 160000 \text{ kg}$$

$$(a) V \text{ (in m}^3\text{)} = ?$$



(b) Load the barge can carry = ?

Solution:

(a)

$$V = w \times l \times h' = (9.15) (26) (3.6) = 856.44 \text{ m}^3$$

(b)

$$\text{Buoyant force} = V \times \rho \times g$$

$$\text{Buoyant force} = (856.44) (1000) (9.8) = 8.4 \times 10^6 \text{ N}$$

$$\text{Weight the barge can carry} = \text{Buoyant force} - \text{Weight of empty barge}$$

$$\text{Weight the barge can carry} = 8.4 \times 10^6 - 160000 \times 9.8$$

$$\text{Weight the barge can carry} = 6.832 \times 10^6 \text{ N}$$

$$\text{Load the barge can carry} = \frac{6.832 \times 10^6}{9.8}$$

$$\text{Load the barge can carry} = 6.971 \times 10^5 \text{ kg}$$

$$\text{Load the barge can carry} = \frac{6.971 \times 10^5}{1000}$$

$$\text{Load the barge can carry} = \mathbf{697.1 \text{ tons}}$$

6. A canal lock gate is 20m wide and 10m deep. Calculate the thrust acting on it assuming that the water in the canal is in level with the top of the gate.

Density of water is 1000 g/m³.

Data:

$$w = 20\text{m}$$

$$h = 10\text{m}$$

$$\rho = 1000 \text{ kg/m}^3$$

$$F_{\text{thrust}} = ?$$

Solution:

$$F_{\text{thrust}} = P \times A$$

$$F_{\text{thrust}} = \frac{1}{2} \rho g h \times A$$

$$F_{\text{thrust}} = \frac{1}{2} (1000) (9.8) (10) (10 \times 20)$$

$$F_{\text{thrust}} = \mathbf{9.8 \times 10^6 \text{ N}}$$

7. A tank 4 m long, 3 m wide and 2 m deep is filled to the brim with paraffin (density 800 kg/m³). Calculate the pressure on the base? What is the thrust on the base?

Data:

$$l = 4\text{m}$$

$$w = 3\text{m}$$

$$h = 2\text{m}$$

$$P = ?$$

$$F_{\text{thrust}} = ?$$

Solution:



$$P = \rho gh$$

$$P = (800) (10) (2)$$

$$P = 16000 \text{ Pa}$$

$$F_{\text{thrust}} = P \times A = P \times (l \times w)$$

$$F_{\text{thrust}} = 16000 \times (4 \times 3)$$

$$F_{\text{thrust}} = 192000 \text{ N}$$

8. A rectangular boat is 4.0 m wide, 8.0 m long, and 3.0 m deep. (a) How much water will it displace if the top stays 1.0 m above the water? (b) What load will the boat contain under these conditions if the empty boat weighs $8.60 \times 10^4 \text{ N}$ in dry dock?

(Weight density of water 9800 N/m^3)

Data:

$$w = 4 \text{ m}$$

$$l = 8 \text{ m}$$

$$h = 3 \text{ m}$$

$$h' = 3 - 1 = 2 \text{ m}$$

$$\text{Weight of empty boat} = 8.60 \times 10^4 \text{ N}$$

(a) $V = ?$

(b) Weight the boat can carry = ?

Solution:

(a)

$$V = l \times w \times h' = 8 \times 4 \times 2 = 64 \text{ m}^3$$

(b)

$$\text{Buoyant force} = V \times \rho \times g$$

$$\text{Buoyant force} = (64) (1000) (9.8) = 6.272 \times 10^5 \text{ N}$$

$$\text{Weight the boat can carry} = \text{Buoyant force} - \text{Weight of empty boat}$$

$$\text{Weight the boat can carry} = 6.272 \times 10^5 - 8.60 \times 10^4 = 5.412 \times 10^5 \text{ N}$$

9. A hot air balloon has a volume of 2200 m^3 . The density of air at temperature of 20°C is 1.205 kg/m^3 . The density of the hot air inside the balloon at a temperature of 100°C is 0.946 kg/m^3 . How much weight can the hot air can lift?

Data:

$$V = 2200 \text{ m}^3$$

$$\rho_{\text{cold}} = 1.205 \text{ kg/m}^3$$

$$\rho_{\text{hot}} = 0.946 \text{ kg/m}^3$$

$$W = ?$$

Solution:

$$W = (V \times \rho_{\text{cold}} - V \times \rho_{\text{hot}}) \times g$$

$$W = (2200 \times 1.205 - 2200 \times 0.946) \times 9.8$$

$$W = 5584.04 \text{ N}$$



10. A spherical balloon has a radius of 7.15 m and is filled with helium. How large a cargo can it lift, assuming that the skin and structure of the balloon have a mass of 930 kg? Neglect the buoyant force on the cargo volume itself.

Data:

$$r = 7.15 \text{ m}$$

$$\rho_{\text{He}} = 0.179 \text{ kg/m}^3$$

$$\rho_{\text{air}} = 1.29 \text{ kg/m}^3$$

$$m_{\text{balloon}} = 930 \text{ kg}$$

$$m_{\text{cargo}} = ?$$

$$W_{\text{cargo}} = ?$$

Solution:

$$V = \pi r^3$$

$$V = \pi (7.15)^3$$

$$V = 1531.111 \text{ m}^3$$

$$F = \rho_{\text{air}} V g$$

$$F = (1.29) (1531.111) (9.8)$$

$$F = 1.94 \times 10^4 \text{ N}$$

$$F = 1.98 \times 10^4 \text{ N}$$

$$m_{\text{He}} = \rho_{\text{He}} V$$

$$m_{\text{He}} = (0.179) (1531.111)$$

$$m_{\text{He}} = 274.1 \text{ kg}$$

$$F = m_{\text{cargo}} g + m_{\text{balloon}} g + m_{\text{He}} g$$

$$1.94 \times 10^4 = m_{\text{cargo}} (9.8) + (930) (9.8) + (274.1) (9.8)$$

$$m_{\text{cargo}} = 775.5 \text{ kg}$$

$$W_{\text{cargo}} = m_{\text{cargo}} g$$

$$W_{\text{cargo}} = 775.5 \times 9.8$$

$$W_{\text{cargo}} = 7.6 \times 10^3 \text{ N}$$



Chapter 7

Numericals:

1. Two spherical raindrops of equal size are falling through air at a velocity of 0.08 m/s. If the drops join together forming a large spherical drop, what will be the new terminal velocity?

Data:

Velocity of small drops = 0.08 m/s

Velocity of big drop = ?

Solution:

Terminal velocity of the big drop = $2^{\frac{2}{3}}$ Velocity of each small drop

Terminal velocity of the big drop = $0.08 \times 2^{\frac{2}{3}}$

Terminal velocity of the big drop = 0.13 m/s

2. Calculate the viscous drag on a drop of oil of 0.1 mm radius falling through air at this terminal velocity. (Viscosity of air = 1.8×10^{-5} Pa.s; density of oil = 850 kg/m³).

Data:

$F = ?$

$r = 0.1 \text{ mm} = 0.1 \div 1000 = 0.0001 \text{ m}$

$\eta = 1.8 \times 10^{-5} \text{ Pa.s}$

$\rho = 850 \text{ kg/m}^3$

Solution:

$v = 2r^2 \times (\rho - \rho_0) \times g / 9\eta$

$v = 2(0.0001)^2 \times (850 - 0) \times 9.8 / 9(1.8 \times 10^{-5})$

$v = 1.03 \text{ m/s}$

$F = 6\pi\eta rv$

$F = 6\pi(1.8 \times 10^{-5})(0.0001)(1.03)$

$F = 3.495 \times 10^{-8} \text{ N}$

3. What area must a heating duct have if air moving 3.0 m/s along it can replenish the air every 15 minutes in a room of volume 300m³. Assume air density remains constant.

Data:

$A = ?$

$\Delta v = 3.0 \text{ m/s}$

$\Delta t = 15 \text{ mins} = 15 \times 60 = 900 \text{ s}$

$V = 300 \text{ m}^3$

Solution:



$$\frac{\Delta v}{\Delta t} = A v$$
$$\frac{300}{900} = A$$

$$A = 0.111 \text{ m}^2$$

4. Water circulates throughout a house in a hot-water heating system. If the water is pumped at a speed of 0.50 m/s through a 4.0 cm diameter pipe in the basement under a pressure of 3.0 atm, what will be the flow speed and pressure in a 2.6 cm diameter pipe on the second floor 5.0 m above? Assume the pipes do not divide into branches.

Data:

$$v_1 = 0.50 \text{ m/s}$$

$$d_1 = 4.0 \text{ cm} = 4 \div 100 = 0.04 \text{ m}$$

$$P_1 = 3.0 \text{ atm} = 3 \times 10^5 \text{ Pa}$$

$$v_2 = ?$$

$$P_2 = ?$$

$$d_2 = 2.6 \text{ cm} = 2.6 \div 100 = 0.026 \text{ m}$$

$$h_2 = 5 \text{ m}$$

Solution:

$$A_1 = \frac{\pi}{4} \times d_1^2$$

$$A_1 = \frac{\pi}{4} \times (0.04)^2$$

$$A_1 = 1.26 \times 10^{-3} \text{ m}^2$$

$$A_2 = \frac{\pi}{4} \times d_2^2$$

$$A_2 = \frac{\pi}{4} \times (0.026)^2$$

$$A_2 = 5.3 \times 10^{-4} \text{ m}^2$$

Since flow rate is same

$$A_1 v_1 = A_2 v_2$$

$$(1.26 \times 10^{-3}) (0.5) = (5.3 \times 10^{-4}) v_2$$

$$v_2 = 1.2 \text{ m/s}$$

$$p_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = p_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

$$3 \times 10^5 + \frac{1}{2} (1000) (0.50)^2 + (1000) (9.8) (0) = p_2 + \frac{1}{2} (1000) (1.2)^2 + (1000) (9.8) (5)$$

$$p_2 = 250405 \text{ Pa}$$

$$p_2 = 250405 \div 1000$$

$$p_2 = 2.5 \text{ atm}$$



5. What is the volume rate of flow of water from a 1.85 cm diameter faucet if the pressure head is 12 m?

Data:

$$d = 1.85 \text{ cm} = 1.85 \div 100 = 0.0185 \text{ m}$$

$$h = 12 \text{ m}$$

$$\frac{\Delta v}{\Delta t} = ?$$

Solution:

$$A = \frac{\pi}{4} \times d^2$$

$$A = \frac{\pi}{4} \times 0.0185^2$$

$$A = 2.7 \times 10^{-4} \text{ m}^2$$

$$v = \sqrt{2gh}$$

$$V = \sqrt{2(9.8)(12)}$$

$$V = 15.34 \text{ m/s}$$

$$\frac{\Delta v}{\Delta t} = A v$$

$$\frac{\Delta v}{\Delta t} = (2.7 \times 10^{-4}) (15.34) = 4.142 \times 10^{-3} \text{ m}^3/\text{s}$$

6. The stream of water emerging from a faucet 'neck down' as it falls. The cross sectional area is 1.2 cm² and 0.35 cm². The two levels are separated by a vertical distance of 45 mm as shown in figure. At what rate does water flow from the tap?

Data:

$$A_0 = 1.2 \text{ cm}^2 = 1.2 \div 10000 = 0.00012 \text{ m}^2$$

$$A = 0.35 \text{ cm}^2 = 0.35 \div 10000 = 0.000035 \text{ m}^2$$

$$h = 45 \text{ mm} = 45 \div 1000 = 0.045 \text{ m}$$

$$\frac{\Delta v}{\Delta t} = ?$$

Solution:

At both levels pressure is atmospheric pressure

$$p_1 = p_2 = p_{\text{atm}}$$

$$p_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = p_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

$$p_{\text{atm}} + \frac{1}{2} \rho v^2 + \rho g(0) = p_{\text{atm}} + \frac{1}{2} \rho v_0^2 + \rho g h$$

Since flow rate is same

$$A_0 v_0 = A v$$

$$\frac{A_0 v_0}{A} = v$$

$$\frac{1}{2} \rho \left(\frac{A_0 v_0}{A} \right)^2 = \frac{1}{2} \rho v_0^2 + \rho g h$$

$$\frac{1}{2} (1000) \left(\frac{0.00012 v_0}{0.000035} \right)^2 = \frac{1}{2} (1000) v_0^2 + (1000) (9.8) (0.045)$$

$$5877.551 v_0^2 = 500 v_0^2 + 441$$



$$v_0 = 0.29 \text{ m/s}$$

$$\frac{\Delta v}{\Delta t} = A_0 v_0$$

$$\frac{\Delta v}{\Delta t} = (0.00012) (0.29)$$

$$\frac{\Delta v}{\Delta t} = 3.48 \times 10^{-5} \text{ m}^3/\text{s}$$

$$\frac{\Delta v}{\Delta t} = 3.48 \times 10^{-5} \times 1 \times 10^6$$

$$\frac{\Delta v}{\Delta t} = 34.8 \text{ cm}^3/\text{s}$$

7. Water leaves the jet of a horizontal hose at 10 m/s. If the velocity of water within the hose is 0.40 m/s, calculate the pressure within the hose. Density of water is 1000 kg/m³ and atmospheric pressure is 100000 Pa.

Data:

$$v_2 = 10 \text{ m/s}$$

$$v_1 = 0.40 \text{ m/s}$$

$$p_1 = ?$$

$$p_2 = 100000 \text{ Pa}$$

$$\rho = 1000 \text{ kg/m}^3$$

Solution:

Assuming both points are at the same level

$$h_1 = h_2 = 0$$

$$p_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = p_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

$$p_1 + \frac{1}{2} (1000) (0.4)^2 + \rho g (0) = (100000) + \frac{1}{2} (1000) (10)^2 + \rho g (0)$$

$$p_1 = 1.5 \times 10^5 \text{ Pa}$$

8. What is the maximum weight of an aircraft with a wing area of 50 m² flying horizontally, is the velocity of the air over the upper surface of the wing is 150 m/s and that the lower surface 140 m/s? Density of air is 1.29 kg/m³.

Data:

$$A = 50 \text{ m}^2$$

$$v_1 = 140 \text{ m/s}$$

$$v_2 = 150 \text{ m/s}$$

$$\rho = 1.29 \text{ kg/m}^3$$

$$W = ?$$

Solution:

There is no too much difference between the points therefore we are going to neglect heights

$$h_1 = h_2 = 0$$

$$p_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = p_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$



$$p_1 + \frac{1}{2} \rho v_1^2 + \rho g(0) = p_2 + \frac{1}{2} \rho v_2^2 + \rho g(0)$$

$$p_1 + \frac{1}{2} \rho v_1^2 = p_2 + \frac{1}{2} \rho v_2^2$$

$$p_1 - p_2 = \Delta p$$

$$\Delta p = \frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2$$

$$\Delta p = \frac{1}{2} (1.29) (150)^2 - \frac{1}{2} (1.29) (140)^2$$

$$\Delta p = 1870.5 \text{ Pa}$$

$$F = W = \Delta p A$$

$$W = (1870.5)(50)$$

$$W = 9.35 \times 10^4 \text{ N}$$

9. A liquid flows through a pipe with a diameter of 0.50 m at a speed of 4.20 m/s. What is the rate of flow in L/min?

Data:

$$d = 0.50 \text{ m}$$

$$v = 4.20 \text{ m/s}$$

$$\frac{\Delta v}{\Delta t} = ? \text{ (in L/min)}$$

Solution:

$$A = \frac{\pi}{4} \times d^2$$

$$A = \frac{\pi}{4} \times (0.5)^2$$

$$A = 0.196 \text{ m}^2$$

$$A = Av$$

$$A = (0.196) (4.20)$$

$$A = 0.8232 \text{ m}^3/\text{s}$$

$$A = 0.8232 \times 1000 \times 60$$

$$A = 49392 \text{ L/min}$$

10. Calculate the average speed of blood flow in the major arteries of the body, which have a total cross-sectional area of about 2.1 cm². Use the data of example. (Take = 28.56 cm³/s)

Data:

$$A = 2.1 \text{ cm}^2$$

$$\frac{\Delta v}{\Delta t} = 28.56 \text{ cm}^3/\text{s}$$

$$v = ?$$

Solution:

$$\frac{\Delta v}{\Delta t} = Av$$



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$$28.56 = (2.1) v$$

$$v = 13.6 \text{ cm/s}$$

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Chapter 8

Numericals:

1. (a) Calculate the value of two equal charges if they repel one another with a force of 0.1 N when situated 50 cm apart in a vacuum.

(b) What would be the size of the charges if they were situated in an insulating liquid whose permittivity was ten times that of a vacuum?

Data:

(a) $F = 0.1 \text{ N}$

$r = 50 \text{ cm} = 0.5 \text{ m}$

$k = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$

$q_1 = q_2 = q$

(b) $F' = 0.1 \text{ N}$

$q' = ?$, when $\epsilon = 10\epsilon_0$

Solution:

(a)

$$F = \frac{kq_1q_2}{r^2}$$

$$0.1 = 9 \times 10^9 \times \frac{q^2}{0.5^2}$$

$$q = 1.7 \times 10^{-6} \text{ C}$$

(b)

$$F' = \frac{F}{\epsilon}$$

$$0.1 = \frac{1}{10} 9 \times 10^9 \times \frac{q'^2}{0.5^2}$$

$$q' = 5.3 \times 10^{-6} \text{ C}$$

2. How far apart must two protons be if the magnitude of the electrostatic force acting on either one due to the other is equal to the magnitude of the gravitational force on a proton at Earth's surface? (mass of proton = $1.67 \times 10^{-27} \text{ kg}$, charge of proton = $1.6 \times 10^{-19} \text{ C}$)

Data:

$r = ?$

$F_g = F_e$

$q = 1.6 \times 10^{-19} \text{ C}$

$m = 1.67 \times 10^{-27} \text{ kg}$

Solution:

$F_g = F_e$

$mg = kq^2/r^2$



$$(1.67 \times 10^{-27}) (9.8) = \frac{8.988 \times 10^9 (1.6 \times 10^{-19})^2}{r^2}$$

$$r = 0.119 \text{ m}$$

3. An electron of charge $1.6 \times 10^{-19} \text{ C}$ is situated in a uniform electric field of intensity $1200 \text{ volt cm}^{-1}$. Find the force on it, its acceleration, and the time it takes to travel 2 cm from rest (electronic mass, $m = 9.1 \times 10^{-31} \text{ kg}$).

Data:

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$E = 1200 \text{ volt cm}^{-1} = 120000 \text{ V/m}$$

$$F = ?$$

$$a = ?$$

$$t = ?$$

$$s = 2 \text{ cm} = 0.02 \text{ m}$$

$$v_i = 0$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

Solution:

$$E = \frac{F}{q}$$

$$F = Eq$$

$$F = 120000 \times 1.6 \times 10^{-19}$$

$$F = 1.92 \times 10^{-14} \text{ N}$$

$$F = ma$$

$$1.92 \times 10^{-14} = (9.1 \times 10^{-31}) a$$

$$a = 2.11 \times 10^{16} \text{ m/s}^2$$

$$S = v_i t + \frac{1}{2} a t^2$$

$$0.02 = (0) t + \frac{1}{2} (2.11 \times 10^{16}) t^2$$

$$t = 1.37 \times 10^{-9} \text{ s}$$

4. An alpha particle (the nucleus of a helium atom) has a mass of $6.64 \times 10^{-27} \text{ kg}$ and a charge of $2e$. What are the (a) magnitude and (b) direction of the electric field that will balance the gravitational force on the particle?

Data:

$$m = 6.64 \times 10^{-27} \text{ kg}$$

$$q = 2e = 2(1.6 \times 10^{-19}) = 3.2 \times 10^{-19} \text{ C}$$

$$(a) E = ?$$

$$(b) \text{ Direction of } E = ?$$

Solution:



(a)

$$F_g = mg$$

$$F_g = (6.64 \times 10^{-27}) (9.8)$$

$$F_g = 6.5 \times 10^{-26} \text{ N}$$

$$E = \frac{F_g}{q}$$

$$E = \frac{6.5 \times 10^{-26}}{3.2 \times 10^{-19}}$$

$$E = 2.03 \times 10^{-7} \text{ N/C}$$

(b)

Since the force of gravity is downward, then the force due to the electric field must point upward. Since $q > 0$ in this situation, this implies the electric field must itself point upward.

5. A proton and an electron form two corners of an equilateral triangle of side length $2.0 \times 10^{-6} \text{ m}$. What is the magnitude of the net electric field these two particles produce at the third corner?

Data:

$$r = 2.0 \times 10^{-6} \text{ m}$$

$$E_{\text{net}} = ?$$

Solution:

From the figure below we see that,

$$E_e = E_p = \frac{e}{Kr^2}$$

We note that the components along the y axis cancel during the vector summation.

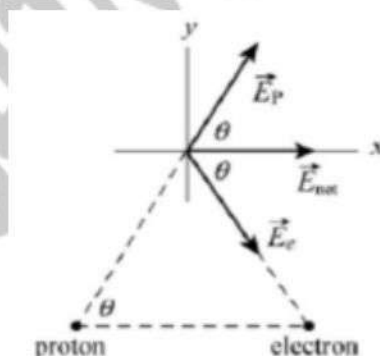
With $\theta = 60^\circ$, the magnitude of the net electric field is obtained as follows:

$$E_{\text{net}} = E_x = 2E_e \cos\theta$$

$$E_{\text{net}} = 2 \left(\frac{e}{Kr^2} \right) \cos\theta$$

$$E_{\text{net}} = 2 \times \frac{1.6 \times 10^{-19}}{9 \times 10^9 \times (2 \times 10^{-6})^2} \cos 60$$

$$E_{\text{net}} = 3.6 \times 10^2 \text{ N/C}$$



6. Figure shows two charged particles on an x axis: $-q = -3.20 \times 10^{-19} \text{ C}$ at $x = -3.00 \text{ m}$ and $q = 3.2 \times 10^{-19} \text{ C}$ at $x = 3.00 \text{ m}$. What are the (a) magnitude and (b) direction (relative to the positive direction of the x axis) of the net electric field produced at point P at $y = 4.00 \text{ m}$?

Data:



$$-q = -3.20 \times 10^{-19} \text{ C}$$

$$x = -3.00 \text{ m}$$

$$q = 3.2 \times 10^{-19} \text{ C}$$

$$x = 3.00 \text{ m}$$

$$E_{\text{net}} = ?$$

$$\text{Direction} = ?$$

Solution:

From the figure below we see that,

$$E = \frac{q}{kr^2}$$

$$\tan \theta = \frac{4}{3}$$

$$\theta = 53.13^\circ$$

$$r = \sqrt{3^2 + 4^2}$$

$$r = 5 \text{ m}$$

We note that the components along the y axis cancel during the vector summation.

With $\theta = 53.13^\circ$, the magnitude of the net electric field is obtained as follows:

$$E_{\text{net}} = E_x = 2E \cos \theta$$

$$E_{\text{net}} = 2 \frac{q}{kr^2} \cos \theta$$

$$E_{\text{net}} = 2 \times \frac{1.6 \times 10^{-19}}{9 \times 10^9 \times (5 \times 10^{-6})^2} \times \cos 53.13^\circ$$

$$E_{\text{net}} = 1.38 \times 10^{-10} \text{ N/C}$$

The net electric field points in the $-x$ direction

7. A proton and an electron form two corners of an equilateral triangle of side length $5 \mu\text{m}$. What is the magnitude of the net electric field these two particles produce at the third corner?

Data:

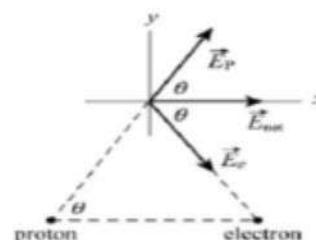
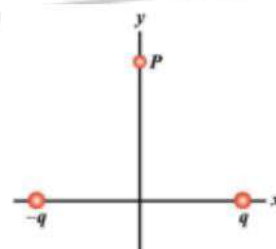
$$r = 5 \mu\text{m} = 5 \times 10^{-6} \text{ m}$$

$$E_{\text{net}} = ?$$

Solution:

From the figure below we see that,

$$E_e = E_p = \frac{e}{kr^2}$$





We note that the components along the y axis cancel during the vector summation.
With $\theta=60^\circ$, the magnitude of the net electric field is obtained as follows:

$$E_{\text{net}} = E_x = 2 \frac{e}{kr^2} \cos\theta$$

$$E_{\text{net}} = 2 \times \frac{1.6 \times 10^{-19}}{9 \times 10^9 \times (5 \times 10^{-6})^2} \times \cos 60^\circ$$

$$E_{\text{net}} = 57.52 \text{ N/C}$$

8. The square surface shown in figure measures 3.2 mm on each side. It is immersed in a uniform electric field with magnitude $E=1800 \text{ N/C}$ and with field lines at an angle of 35° with a normal to the surface, as shown. Take that normal to be directed "outward," as though the surface were one face of a box. Calculate the electric flux through the surface.

Data:

$$l = 3.2 \text{ mm} = 0.0032 \text{ m}$$

$$E = 1800 \text{ N/C}$$

$$\theta' = 35^\circ$$

$$\theta = 180^\circ - 35^\circ = 145^\circ$$

$$\phi_e = ?$$

Solution:

$$A = l^2 = (0.0032)^2 = 1.024 \times 10^{-5} \text{ m}^2$$

$$\phi_e = EA \cos \theta$$

$$\phi_e = (1800) (1.024 \times 10^{-5}) (\cos 145^\circ)$$

$$\phi_e = -1.5 \times 10^{-2} \text{ N.m}^2/\text{C}$$

9. An electron is liberated from the lower of two large parallel metal plates separated by a distance $h = 2\text{cm}$. The upper plate has a potential of 2400 volts relative to the lower. How long does the electron take to reach it?

Data:

$$h = 2\text{cm} = 0.02 \text{ m}$$

$$V = 2400 \text{ V}$$

Solution:

$$S = v_i t + \frac{1}{2} a t^2$$

$$\text{Since, } v_i = 0$$

$$h = \frac{1}{2} a t^2 \quad \text{eq(1)}$$

$$a = \frac{F}{m} \quad \text{eq (2)}$$

$$F = Eq \quad \text{eq (3)}$$

$$E = \frac{V}{h}$$



Put the value of E in equation (3)

$$F = \frac{Vq}{h}$$

Put in equation (2)

$$a = \frac{Vq}{hm}$$

Put in equation (1)

$$h = \frac{1}{2} \frac{Vq}{hm} t^2$$

$$0.02 = \frac{1}{2} \frac{2400 \cdot 1.6 \times 10^{-19}}{0.02 \times 9.11 \times 10^{-27}} t^2$$

$$t = 1.4 \times 10^{-9} \text{ s}$$

10. Two large parallel metal plates are 1.5 cm apart and have charges of equal magnitudes but opposite signs on their facing surfaces. Take the potential of the negative plate to be zero. If the potential halfway between the plates is then 5.0 V, what is the electric field in the region between the plates?

Data:

$$2d = 1.5 \text{ cm}$$

$$d = 0.75 \text{ cm} = 0.0075 \text{ m}$$

$$V = 5.0 \text{ V}$$

$$E = ?$$

Solution:

$$E = \frac{V}{d}$$

$$E = \frac{5}{0.0075}$$

$$E = 6.7 \times 10^2 \text{ V/m}$$

**Chapter # 9****Numericals:**

1. The capacitance of air-filled parallel plate capacitor is 1.3 pF. If the separation of the plates is doubled and wax is inserted between them. The new capacitance is 2.6 pF. Find the dielectric constant of the wax.

Data:

$$C = 1.3 \text{ pF}$$

$$C' = 2.6 \text{ pF}$$

$$\epsilon_r = ?$$

Solution:

$$C = \frac{\epsilon_0 A}{d}$$

$$C' = \frac{\epsilon_0 \epsilon_r A}{2d}$$

Dividing C' by C

$$\frac{C'}{C} = \frac{\epsilon_r}{2}$$

$$\epsilon_r = 2 \times \frac{2.6}{1.3}$$

$$\epsilon_r = 4$$

2. Three capacitors have capacitances 10, 50 and 25 μF respectively as shown in figure. Calculate (i) charge on each when connected in parallel to a 250 V supply (ii) total capacitance and (iii) potential difference across each when connected in series.

Data:

$$C_1 = 10 \mu\text{F}$$

$$C_2 = 50 \mu\text{F}$$

$$C_3 = 25 \mu\text{F}$$

Parallel Connection

$$V = 250 \text{ V}$$

$$Q_1 = ?, Q_2 = ?, Q_3 = ?$$

$$C_{eq} = ?$$

Series Connection

$$V_{eq} = 250 \text{ V}$$

$$V_1 = ?, V_2 = ?, V_3 = ?$$

Solution:

(i)

$$Q_1 = C_1 V = (10 \times 10^{-6}) (250) = 2.5 \times 10^{-3} \text{ C} = 2.5 \text{ mC}$$

$$Q_2 = C_2 V = (50 \times 10^{-6}) (250) = 12.5 \times 10^{-3} \text{ C} = 12.5 \text{ mC}$$

$$Q_3 = C_3 V = (25 \times 10^{-6}) (250) = 6.25 \times 10^{-3} \text{ C} = 6.25 \text{ mC}$$

(ii)

$$C_{eq} = C_1 + C_2 + C_3$$

$$C_{eq} = 10 + 50 + 25 = 85 \mu\text{F}$$



(iii)

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_{eq}} = \frac{1}{10} + \frac{1}{50} + \frac{1}{25}$$

$$\frac{1}{C_{eq}} = \frac{4}{25}$$

$$C_{eq} = 6.25 \mu F$$

$$Q = C_{eq} V_{eq}$$

$$Q = (6.25 \times 10^{-6}) (250)$$

$$Q = 1.5625 \times 10^{-3} \text{ C}$$

$$Q = 1.5625 \text{ mC}$$

$$V_1 = \frac{Q}{C_1}$$

$$V_1 = \frac{1.5625 \times 10^{-3}}{10 \times 10^{-6}}$$

$$V_1 = 156.25 \text{ V}$$

$$V_1 = \frac{1.5625 \times 10^{-3}}{10 \times 10^{-6}}$$

$$V_2 = \frac{Q}{C_2}$$

$$V_2 = \frac{1.5625 \times 10^{-3}}{50 \times 10^{-6}}$$

$$V_2 = 31.25 \text{ V}$$

$$V_3 = \frac{Q}{C_3}$$

$$V_3 = \frac{1.5625 \times 10^{-3}}{25 \times 10^{-6}}$$

$$V_3 = 62.5 \text{ V}$$

3. Three capacitors are connected through a potential difference of 100 volts as shown in figure. Find the charges and potential difference across each capacitor.

Data:

$$V = 100 \text{ V}$$

$$Q_1 = ?, V_1 = ?$$

$$Q_2 = ?, V_2 = ?$$

$$Q_3 = ?, V_3 = ?$$

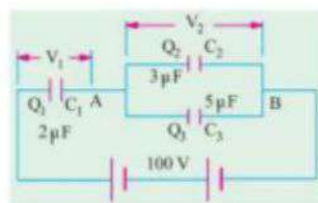
Solution:

Capacitors 2 and 3 are connected in parallel, while their combination is in series with capacitor 1

$$C_A = C_2 + C_3$$

$$C_A = 3 + 5 = 8 \mu F$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_A}$$





$$\frac{1}{C_{eq}} = \frac{1}{2} + \frac{1}{8}$$

$$C_{eq} = 1.6 \mu F$$

$$Q = C_{eq} V$$

$$Q = (1.6 \times 10^{-6}) (100)$$

$$Q = 1.6 \times 10^{-4} C$$

$$Q = 160 \mu C$$

$$Q_1 = Q = 160 \mu C$$

$$V_1 = \frac{Q_1}{C_1}$$

$$V_1 = \frac{160}{2} = 80 V$$

$$Q_A = Q = 160 \mu C$$

$$V_A = \frac{160}{8} = 20 V$$

$$V_2 = V_3 = V_A = 20 V$$

$$Q_2 = C_2 V_2$$

$$Q_2 = (3 \times 10^{-6}) (20)$$

$$Q_2 = 6 \times 10^{-5} C$$

$$Q_2 = 60 \times 10^{-6} C$$

$$Q_2 = 60 \mu C$$

$$Q_3 = C_3 V_3$$

$$Q_3 = (5 \times 10^{-6}) (20)$$

$$Q_3 = 1 \times 10^{-4} C$$

$$Q_3 = 100 \times 10^{-6} C$$

$$Q_3 = 100 \mu C$$

4. Capacitor is charged through a large non-reactive resistance by a battery of constant voltage V. For this arrangement, if the capacitor has a capacitance of 10 μF and the resistance is 1 M Ω , calculate the time taken for the capacitor to receive 90% of its final charge.

Data:

$$C = 10 \mu F$$

$$R = 1 M\Omega$$

$$Q = 0.9Q_0$$

$$t = ?$$

Solution:

$$\tau = RC$$

$$\tau = (1 \times 10^6) (10 \times 10^{-6})$$

$$\tau = 10$$

$$Q = Q_0 (1 - e^{-\frac{t}{\tau}})$$

$$\frac{0.9Q_0}{Q_0} = 1 - e^{-\frac{t}{10}}$$

$$e^{-\frac{t}{10}} = 1 - 0.9$$



$$e^{-\frac{t}{10}} = 0.1$$

Taking ln on both side

$$\ln e^{-\frac{t}{10}} = \ln 0.1$$

$$-\frac{t}{10} \ln e = \ln 0.1$$

$$-\frac{t}{10} = \ln 0.1$$

$$-t = -2.3 \times 10$$

$$t = 23 \text{ s}$$

5. What capacitance is required to store energy of 10 kWh at a potential difference of 1000 V?

Data:

$$E = 10 \text{ kWh} = 10 \times 3.6 \times 10^6 = 3.6 \times 10^7 \text{ J}$$

$$V = 1000 \text{ V}$$

$$C = ?$$

Solution:

$$E = \frac{1}{2} C V^2$$

$$3.6 \times 10^7 = \frac{1}{2} C (1000)^2$$

$$C = 72 \text{ F}$$

6. A 2.0 μF capacitor and a 4.0 μF capacitor are connected in parallel across a 300 V potential difference. Calculate the total energy stored in the capacitors

Data:

$$C_1 = 2.0 \mu\text{F}$$

$$C_2 = 4.0 \mu\text{F}$$

$$V = 300 \text{ V}$$

$$E = ?$$

Solution:

$$C_{\text{eq}} = C_1 + C_2$$

$$C_{\text{eq}} = 2 + 4$$

$$C_{\text{eq}} = 6 \mu\text{F}$$

$$E = \frac{1}{2} C_{\text{eq}} V^2$$

$$E = \frac{1}{2} (6 \times 10^{-6}) (300)^2$$

$$E = 0.27 \text{ J}$$

7. A 12.0 V battery is connected to a capacitor, resulting in 54.0 μC of charge stored on the capacitor. How much energy is stored in the capacitor?

Data:

$$V = 12 \text{ V}$$

$$Q = 54.0 \mu\text{C} = 54 \times 10^{-6} \text{ C}$$

$$E = ?$$

Solution:



$$E = \frac{1}{2} Q V$$

$$E = \frac{1}{2} (54 \times 10^{-6}) (12)$$

$$E = 3.24 \times 10^{-4} \text{ J}$$

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Chapter 10

Numericals:

1. The storage battery of a car has an emf of 12 V. If the internal resistance of the battery is 0.5Ω , what is the maximum current that can be drawn from the battery?

Data:

$$E = 12 \text{ V}$$

$$r = 0.5\Omega$$

$$I = ?$$

Solution:

$$E = Ir$$

$$12 = I (0.5)$$

$$I = 24 \text{ A}$$

2. A negligibly small current is passed through a wire of length 12m and uniform cross-section $4.0 \times 10^{-7} \text{ m}^2$, and its resistance is measured to be 6.0Ω . What is the resistivity of the material at the temperature of the experiment?

Data:

$$L = 12\text{m}$$

$$A = 4.0 \times 10^{-7} \text{ m}^2$$

$$R = 6.0 \Omega$$

$$\rho = ?$$

Solution:

$$R = \frac{\rho L}{A}$$

$$6 = \frac{\rho(12)}{4.0 \times 10^{-7}}$$

$$\rho = 2 \times 10^{-7} \Omega\text{m}$$

3. In a potentiometer arrangement, a cell of emf 1.20 V gives a balance point at 40.0 cm length of the wire. If the cell is replaced by another cell and the balance point shifts to 74.0 cm, what is the emf of the second cell?

Data:

$$E_1 = 1.2 \text{ V}$$

$$L_1 = 40 \text{ cm}$$

$$L_2 = 74 \text{ cm}$$

$$E_2 = ?$$

Solution:

$$\frac{E_1}{E_2} = \frac{L_1}{L_2}$$



$$\frac{E_2}{E_1} = \frac{L_2}{L_1}$$

$$\frac{E_2}{1.2} = \frac{74}{40}$$

$$E_2 = 2.22 \text{ V}$$

4. (a) Three resistors 1Ω , 2Ω , and 3Ω are combined in series. What is the total resistance of the combination?

b) If the combination is connected to a battery of emf 24 V and negligible internal resistance, obtain the potential drop across each resistor.

Data:

(a) $R_1 = 1\Omega$, $R_2 = 2\Omega$, $R_3 = 3\Omega$, series

(b) $V_{eq} = 24 \text{ V}$ (because internal resistance is negligible), $V_1 = ?$, $V_2 = ?$, $V_3 = ?$

Solution:

(a)

$$R_{eq} = R_1 + R_2 + R_3$$

$$R_{eq} = 1 + 2 + 3$$

$$R_{eq} = 6\Omega$$

(b)

$$V_{eq} = IR_{eq}$$

$$24 = I(6)$$

$$I = 4 \text{ A}$$

$$V_1 = IR_1 = (4)(1) = 4\text{V}$$

$$V_2 = IR_2 = (4)(2) = 8\text{V}$$

$$V_3 = IR_3 = (4)(3) = 12\text{V}$$

5. From the given circuit find the value of I .

Data:

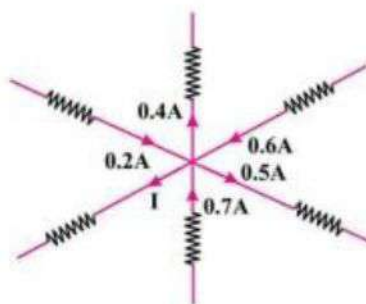
$$I = ?$$

Solution:

$$\Sigma I_{in} = \Sigma I_{out}$$

$$0.2 + 0.6 + 0.7 = 0.4 + 0.5 + I$$

$$I = 0.6 \text{ A}$$



6. In a meter bridge with a standard resistance of 15Ω in the right gap, the ratio of balancing length is $5:3$. Find the value of the other resistance.

Data:



$$\frac{L_1}{L_2} = 5:3$$

$$R = 15\Omega$$

$$X = ?$$

Solution:

$$\frac{X}{R} = \frac{L_1}{L_2}$$

$$\frac{X}{15} = \frac{5}{3}$$

$$X = 25\Omega$$

7. By using KVL find Current flowing through 10Ω resistance.

Data:

$$I_2 = ?$$

Solution:

For left loop

Voltage rise = Voltage drop

$$4 = 5 I_1 + 15(I_1 - I_2)$$

$$20 I_1 - 15 I_2 = 4 \quad \text{eq (1)}$$

For middle loop

Voltage rise = Voltage drop

$$0 = 15(I_2 - I_1) + 10 I_2 + 8 (I_2 - I_3)$$

$$-15 I_1 + 33 I_2 - 8 I_3 = 0$$

$$15 I_1 - 33 I_2 + 8 I_3 = 0 \quad \text{eq (2)}$$

For right loop

Voltage rise = Voltage drop

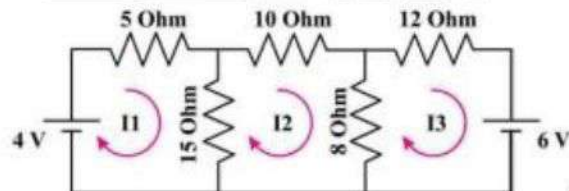
$$-6 = 8 (I_3 - I_2) + 12 I_3$$

$$-8 I_2 + 20 I_3 = -6$$

$$8 I_2 - 20 I_3 = 6 \quad \text{eq (3)}$$

By using calculator

$I_2 = 0.0323 \text{ A}$ (since sign is positive therefore I_2 in clockwise direction)



**Chapter # 11****Numericals:**

1. The period of oscillation of an object in an ideal spring and mass system is 0.50s and the amplitude is 5.0 cm. what is the speed at the equilibrium point? and the acceleration at the point of maximum extension of the spring.

Data:

$$T = 0.50 \text{ s}$$

$$x_{\max} = 5.0 \text{ cm} = 5 \div 100 = 0.05 \text{ m}$$

$$v_{\max} = ?$$

Solution:

$$\omega = \frac{2\pi}{T}$$

$$\omega = \frac{2\pi}{0.5}$$

$$\omega = 12.566 \text{ rad/s}$$

$$v_{\max} = \omega x_{\max}$$

$$v_{\max} = (12.566) (0.05)$$

$$v_{\max} = 0.628 \text{ m/s}$$

$$v_{\max} = 0.628 \times 100$$

$$v_{\max} = 62.8 \text{ cm/s}$$

$$a_{\max} = -\omega^2 x_{\max}$$

$$a_{\max} = -(12.566)^2 (0.05) = -7.9 \text{ m/s}^2$$

The negative sign indicates that acceleration is in opposite direction to displacement.

2. A sewing machine needle moves with a rapid vibratory motion, like SHM, as it sews a seam. Suppose the needle moves 8.4 mm from its highest to its lowest position and it makes 24 stitches in 9.0s. What is the maximum needle speed?

Data:

$$h = 8.4 \text{ mm} = 0.0084 \text{ m}$$

$$\text{No. of stitches} = 24 \text{ stitches}$$

$$t = 9.0 \text{ s}$$

$$v_{\max} = ?$$

Solution:

$$T = \frac{t}{\text{no. of stitches}}$$

$$T = \frac{9}{24}$$



$$T = 0.375 \text{ stitches/s}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.375} = 16.755 \text{ rad/s}$$

$$x_{\max} = \frac{0.0084}{2} = 4.2 \times 10^{-3} \text{ m}$$

$$v_{\max} = \omega x_{\max}$$

$$v_{\max} = (16.755) (4.2 \times 10^{-3})$$

$$v_{\max} = 0.07 \text{ m/s}$$

3. An ideal spring with a spring constant of 15 N/m is suspended vertically. A body of mass 0.60 kg is attached to the unstretched spring and released. (a) What is the extension of the spring when the speed is a maximum? (b) What is the maximum speed?

Data:

$$k = 15 \text{ N/m}$$

$$m = 0.60 \text{ kg}$$

$$(a) x = ?$$

$$(b) v_{\max} = ?$$

Solution:

(a)

$$F = W = k x_{\max}$$

$$mg = k x_{\max}$$

$$x_{\max} = \frac{mg}{K}$$

$$x_{\max} = \frac{(0.6)(9.8)}{15}$$

$$x_{\max} = 0.39 \text{ m}$$

(b)

$$\omega = \sqrt{\frac{K}{m}}$$

$$\omega = \sqrt{\frac{15}{0.6}}$$

$$\omega = 5 \text{ rad/s}$$

$$v_{\max} = \omega x_{\max}$$

$$v_{\max} = (5) (0.39)$$

$$v_{\max} = 1.95 \text{ m} \approx 2 \text{ m}$$

4. A body is suspended vertically from an ideal spring of spring constant 2.5 N/m. the spring is initially in its relaxed position. The body is then released and oscillates about its equilibrium position. The motion is described by $y = (4.0\text{cm}) \sin[(0.70\text{rad/s})t]$. What is the maximum kinetic energy of the body?



Data:

$$k = 2.5 \text{ N/m}$$

$$y = (4.0\text{cm}) \sin[(0.70\text{rad/s})]$$

$$K_{\max} = ?$$

Solution:

$$y = (4.0\text{cm}) \sin[(0.70\text{rad/s})]$$

$$\text{Here } x_0 = 4 \text{ cm} = 4 \div 100 = 0.04 \text{ m}$$

$$K_{\max} = \frac{1}{2} k x_0^2$$

$$K_{\max} = \frac{1}{2} (2.5) (0.04)^2$$

$$K_{\max} = 2 \times 10^{-3} \text{ J}$$

$$K_{\max} = 2 \text{ mJ}$$

5. The period of oscillation of a simple pendulum does not depend on the mass of the bob. By contrast the period of a mass–spring system does depend on mass. Explain the apparent condition.

Ans) Simple Pendulum: The period of a simple pendulum is determined by the length of the pendulum and the acceleration due to gravity (g). The formula for the period (T) of a simple pendulum is given by:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

Where:

- T is the period of oscillation.
- L is the length of the pendulum
- g is the acceleration due to gravity (approximately 9.81 m/s^2 on Earth)

As you can see, the mass of the bob (the object at the end of the pendulum) does not appear in this equation. Therefore, the mass of the bob does not affect the period of the pendulum.

Mass–Spring System: To calculate the period of oscillation (T) for a mass–spring system, you need to consider the mass of the object (m) and the spring constant (k). The formula for the period of a mass–spring system is:

$$T = 2\pi \sqrt{\frac{m}{k}}$$

As you can see, the mass (m) is present in this equation, so the period of oscillation for a mass–spring system depends on the mass of the object attached to the spring.

6. What is the period of a simple pendulum of a 6.0 kg mass oscillating on a 4.0 m long string?

Data:

$$T = ?$$

$$m = 6 \text{ kg}$$



$$L = 4.0 \text{ m}$$

Solution:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$T = 2\pi \sqrt{\frac{4}{9.8}}$$

$$T = 4.01 \text{ s}$$

7. A pendulum of length 75 cm and mass 2.5 kg swings with a mechanical energy of 0.015 J. what is its amplitude?

Data:

$$L = 75 \text{ cm} = 75 \div 100 = 0.75 \text{ m}$$

$$m = 2.5 \text{ kg}$$

$$E = 0.015 \text{ J}$$

Solution:

$$\omega = \sqrt{\frac{g}{L}}$$

$$\omega = \sqrt{\frac{9.8}{0.75}}$$

$$\omega = \sqrt{13.066}$$

$$\omega = 3.614 \text{ rad/s}$$

$$E = \frac{1}{2} m \omega^2 A^2$$

$$0.015 = \frac{1}{2} (2.5) (13.067)^2 A^2$$

$$A = 0.03 \text{ m} = 0.03 \times 100 = 3 \text{ cm}$$

8. A pendulum of length L_1 has a period of $T_1 = 0.950 \text{ s}$. The length of the pendulum is adjusted to a new value L_2 such that $T_2 = 1.00 \text{ s}$ what is the ratio L_2/L_1 .

Data:

$$T_1 = 0.950 \text{ s}$$

$$T_2 = 1.00 \text{ s}$$

$$\frac{L_2}{L_1} = ?$$

Solution:

$$\frac{T_1}{T_2} = 2\pi \sqrt{\frac{L_1}{g}} \div 2\pi \sqrt{\frac{L_2}{g}}$$

$$\frac{T_1}{T_2} = 2\pi \sqrt{\frac{L_1}{g}} \times \frac{1}{2\pi \sqrt{L_2}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{L_1}{L_2}}$$



$$\sqrt{\frac{L_1}{L_2}} = \frac{0.95}{1}$$

$$\sqrt{\frac{L_2}{L_1}} = \frac{1}{0.95}$$

$$\sqrt{\frac{L_2}{L_1}} = 1.05$$

$$\frac{L_2}{L_1} = 1.1025$$

9. A wire is hanging from the top of a tower such that the top is not visible due to darkness. How do you calculate the height of tower?

Ans) Connect some mass with the end of the wire. Take a stop watch and set the pendulum in vibration. Calculate the time period of the pendulum. You may find the time of, say ten vibrations, and divide it by 10 to find the time period of the pendulum. Now equation of the simple pendulum is

$$T = 2\pi \sqrt{\frac{l}{g}}$$

π and g are constants. Use the value of T you just calculated in the above equation and find the length of the wire. As the wire hangs from the top of the tower, it will give you the height of the tower.

Note that l gives you the length of the wire and if the wire is above the ground (and principally it should be) add the remaining part to the length of the wire.

10. The amplitude of oscillation of a pendulum decays by a factor of 20.0 in 120 s. By what factor has its energy decayed in that time.

Data:

$$t = 120 \text{ s}$$

$$A' = \frac{A}{20}$$

Solution:

$$E = \frac{1}{2} m \omega^2 A^2$$

$$E' = \frac{1}{2} m \omega^2 A'^2$$

$$E' = \frac{1}{2} m \omega^2 \left(\frac{A}{20}\right)^2$$

$$E' = \frac{1}{400} \frac{1}{2} m \omega^2 A^2$$

$$E' = \frac{1}{400} E$$

(The energy has decreased by a factor of 400)